FABRIC MANUFACTURING

Part: A

Course Code: FE 351-0723

Weft Knitted Structures

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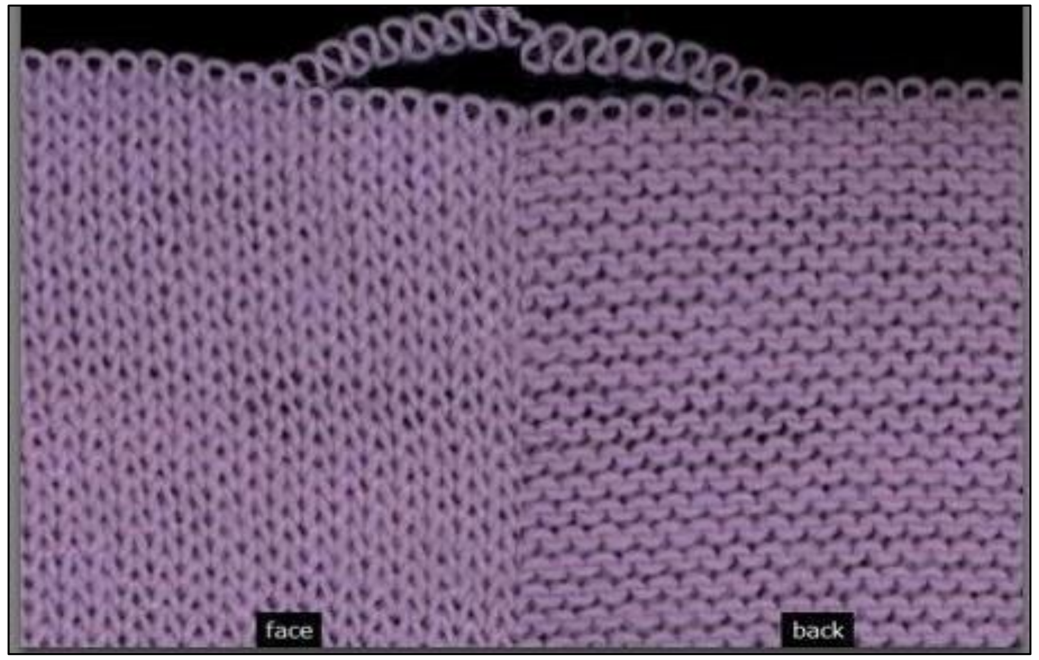
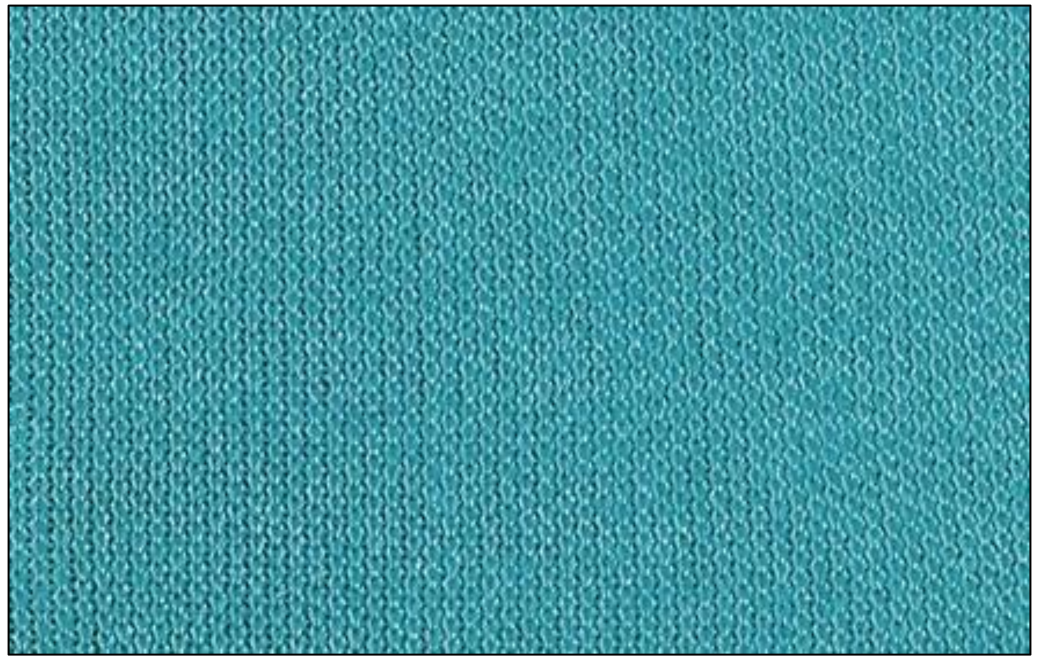
CLASSIFICATIONS OF WEFT KNITTED FABRIC

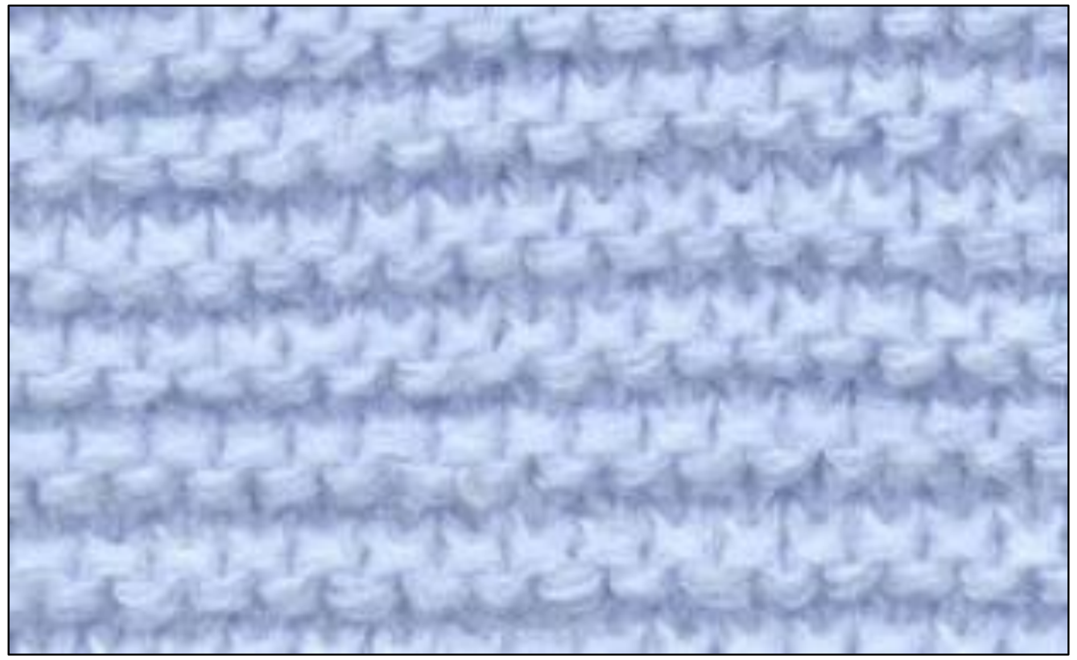
A. Single Jersey Fabric

B. Double Jersey Fabric

1. Rib
2. Interlock
3. Purl



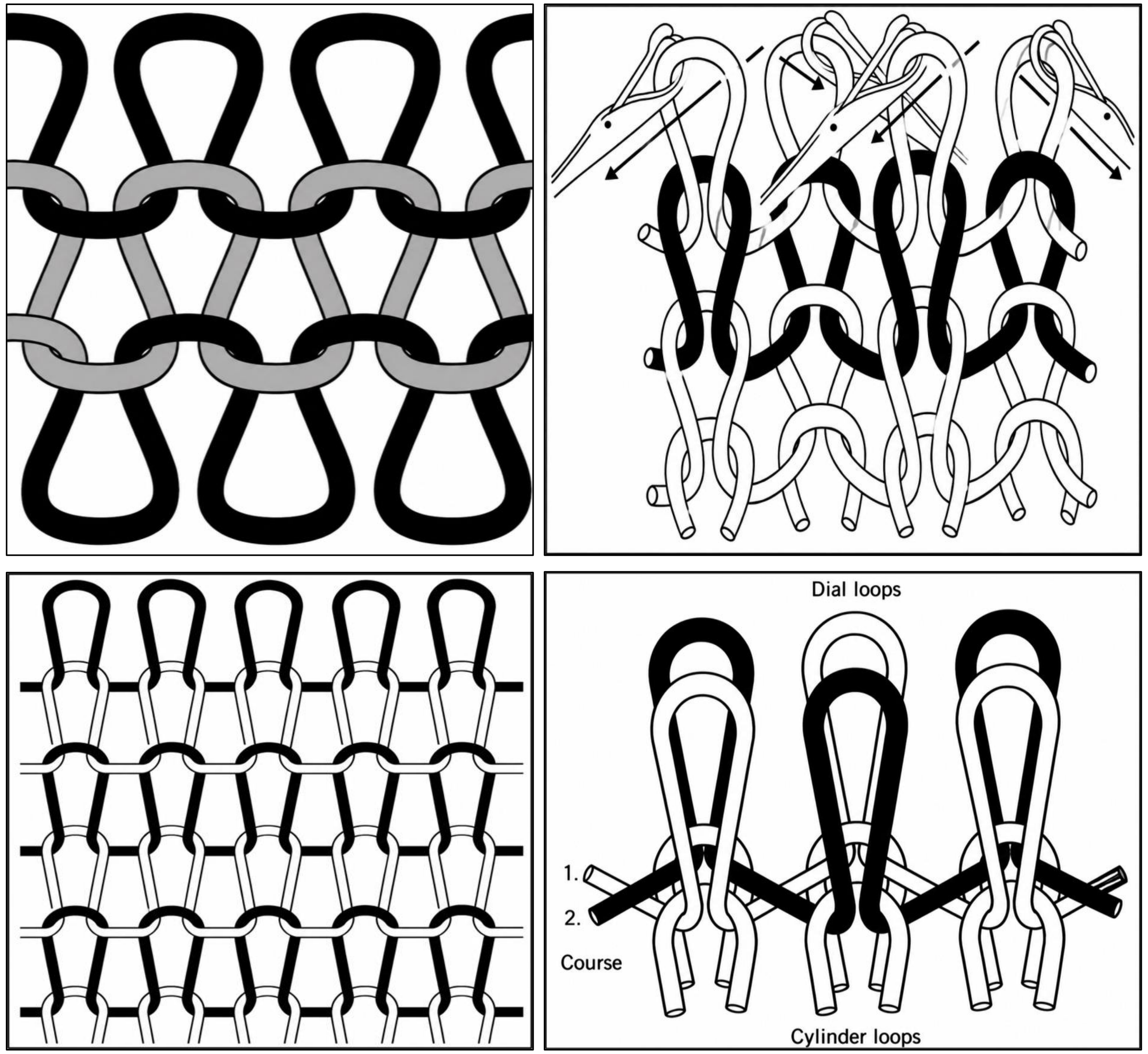




CLASSIFICATIONS OF BASIC WEFT KNITTED STRUCTURE

Basic weft knitted structure:

1. Single Jersey Plain
2. 1x1 Rib
3. 1x1 Purl
4. 1x1 Interlock



https://youtu.be/DP71mEAwb0U?si=7qyfWGze7JoVyJ3Z

FEATURES OF PLAIN FABRIC

- Single jersey plain fabric is simple, most common and economical.
- Both sides are not identical. Each side of the fabric is made of a single type of loop i.e. either face or back (reverse).
- The loops have a V-shaped appearance on the technical face and semi-circular on the technical back of the fabric.
- The V-shaped stitches on the face gives the smooth surface when compared with the back.
- Its production rate is very high because of stitch simplicity.
- The amount of elasticity varies according to the yarn used and the stich length of yarn. The higher the stich length, the more elastic the structure will be and simultaneously the poorer the bursting strength.
- It can be pulled out freely from either end.

FEATURES OF PLAIN FABRIC

1. Appearance

- Single jersey fabric is unbalanced because the two sides look different.
- One side shows face loops and the other side shows back loops.
- The face side is smooth and has a series of V-shaped loops.
- The back side is rough and shows rows of semicircular loops.

2. Extensibility (Stretchability)

- The fabric is stretchable but not always elastic.
- It stretches more in the width direction (course-wise) than in the length direction (wale-wise).
- Course-wise stretch is about twice the wale-wise stretch.
- Recovery after stretching depends on the type of yarn and fiber.

FEATURES OF PLAIN FABRIC

3. Edge Curling

- ❑ Single jersey fabric tends to curl at the edges after cutting.
- ❑ The sides curl towards the back side.
- ❑ The top and bottom edges curl towards the face side.
- ❑ Heat setting and finishing processes are used to reduce curling.

4. Unroving (Unravelling)

- ❑ Plain knit fabric can be easily unravelled from both ends.
- ❑ It can be opened from the first knitted edge and the last knitted edge.
- ❑ This happens because of its simple loop structure.

FEATURES OF PLAIN FABRIC

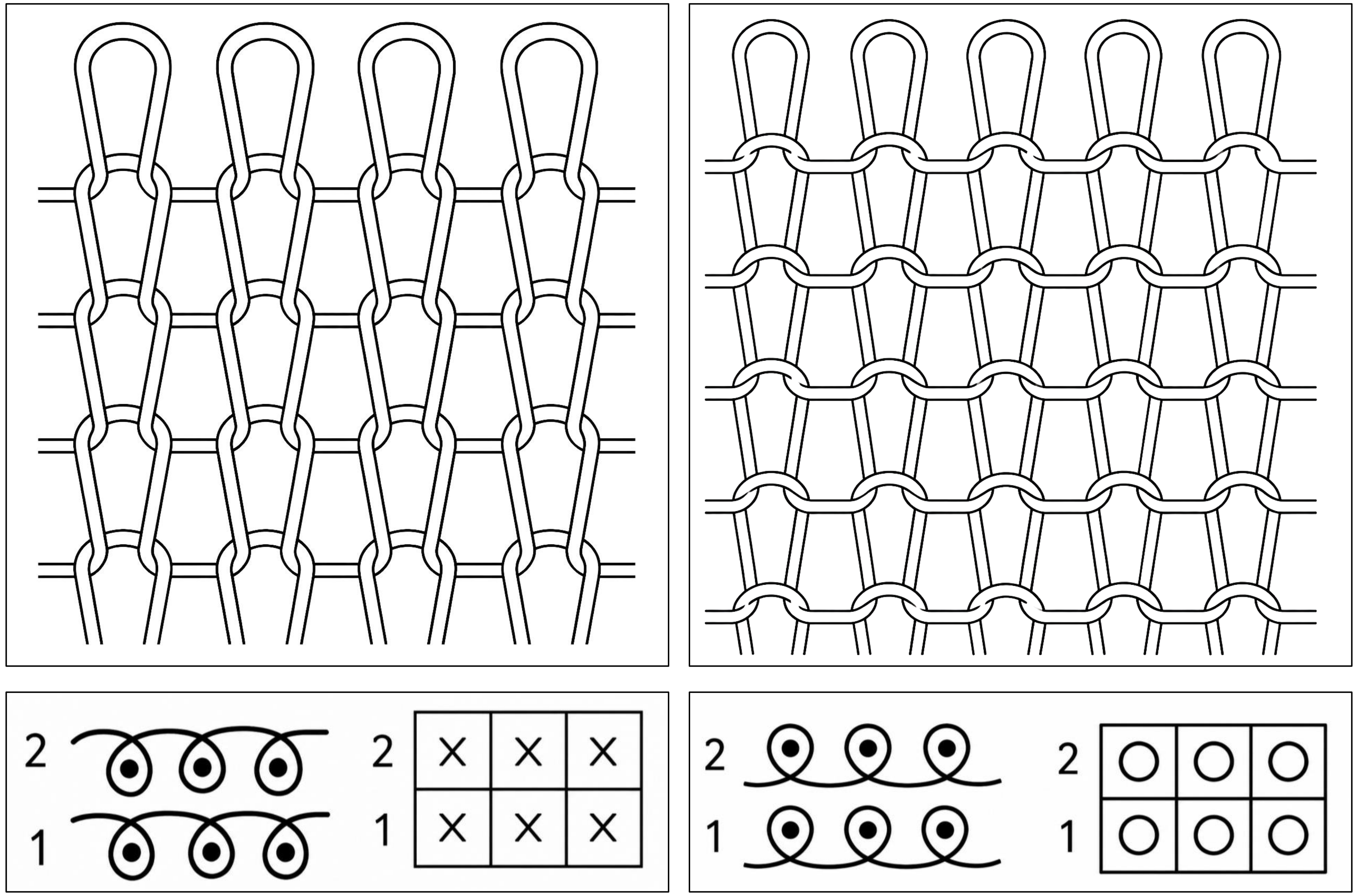
5. Laddering

- If one stitch breaks or drops, the loops in the same wale may fall one after another.
- This creates a vertical line called a ladder.
- Laddering is more common with: Loose fabrics, Smooth yarns.
- It can be repaired manually using a needle.

End uses of plain knit structures:

Plain knit structures are used for basic T-shirt, under garments, men's vest, ladies hosiery, fully-fashioned knitwear etc. End use depends on some factors such as material used, yarn types and yarn count, linear density, machine gauge, fabric thickness and weight, types of finishing etc.

DRAW THE LOOPING DIAGRAM OF TECHNICAL FACE AND TECHNICAL BACK OF A PLAIN FABRIC WITH NOTATION



HOW TO DISTINGUISH A PLAIN KNIT FABRIC?

- Appearance: Different on Face and Back. V shapes on face & arcs on back.
- Unroving: Either end.
- Curling: Tendency to curl.
- Thickness & Warmth: Thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Moderate – High – Mod-High.
- End uses: Ladies stockings, Fine cardigans, Men's and ladies' shirts, Dresses.

FEATURES OF RIB FABRIC

- The simplest rib fabric is 1x1 rib.
- It is knitted with two sets of single ended latch needles.
- Face and back of 1x1 rib are identical. Wales (technical face) prominent on the both sides and the structure can be named as double face or double-knit structure. But drop needle structures (2x2, 3x2, 5x4 etc.) are different.
- The fabric being in many cases symmetrical on both faces is not exposed to unbalanced stress and therefore does not curl, it lies flat, when cut.
- Rib machine requires finer yarn than similar gauge plain machine. But It is heavier structure than plain fabric.
- It is more expensive fabric to produce than plain.
- 1x1 rib is balanced by alternate wale of face loops on both sides. In face side, 1 wales face and another wales back. In back side, 1 wale back, another wale face. So, both face and back loops stay in the same course.
- Both side of the fabric imitate face side of single jersey as back loops remain hidden inside the structure. But back loops can be seen when stretched. A face wales are made by the needles of the cylinder, and the back wales are made by the needles of the dial.

FEATURES OF RIB FABRIC

- Rib structures have alternate gaiting.
- The 1×1 rib fabric is very elastic and can return to its original width after stretching. This happens because the face loop wales move over the reverse loop wales, allowing the fabric to recover easily. Rib fabric is very flexible and spring-like in the width direction because the yarn tries to return to its original shape. Due to this elasticity, rib structures are commonly used for elastic bands in garments, such as cuffs, waistbands, and collars. In normal constructions, rib fabric can stretch up to 120% along the course direction. In the wale direction, rib fabric behaves like plain fabric and has very limited stretchability.
- As the number of wales in each rib increases, the elasticity decreases because the number of changeovers from back to front diminishes. Over 3x3 rib the fabric more and more behaves like plain fabric.
- It cannot be pulled out freely from either ends. It can only be pulled out from edge knitted last. Rib structures, however, cannot be unravelled from the edge knitted first i.e. from the bottom. The arms of the connecting loops enter the loops above them from both sides. Any attempt to pull the yarn causes the connecting loops to tighten and blocks unravelling. Because rib cannot be unravelled from the end knitted first and because of their elasticity, they are suited to the edge of garments such as tops of socks, cuffs and the waist edge of garments.

FEATURES OF RIB FABRIC

1. Appearance

- Rib fabric has a similar appearance on both sides.
- Both sides show face loops because needles from both beds are used.
- The fabric looks like a double face or double-knit fabric.
- Each wale contains only one type of loop (face or back).
- The structure is symmetrical, so it has balanced stress and does not easily curl.

2. Extensibility (Stretchability)

- Rib fabric is highly elastic, especially in the width direction.
- It can stretch and return to its original shape due to its loop structure.
- It is commonly used for cuffs, waistbands and sock tops.
- Lengthwise stretch is limited, similar to plain knit fabric.
- Wider ribs (2×2, 3×3) have less elasticity than 1×1 rib.

FEATURES OF RIB FABRIC

3. Edge Curling

- Rib fabric is symmetrical on both sides.
- It has balanced forces, so it does not curl easily.
- When cut, it usually lies flat.

4. Unroving (Unravelling)

- Rib fabric can be unravelled from the last knitted edge (top edge).
- It cannot be unravelled easily from the first knitted edge (bottom edge).
- Because of this property and its elasticity, rib fabric is suitable for: Socks, Cuffs, Waistbands.

FEATURES OF RIB FABRIC

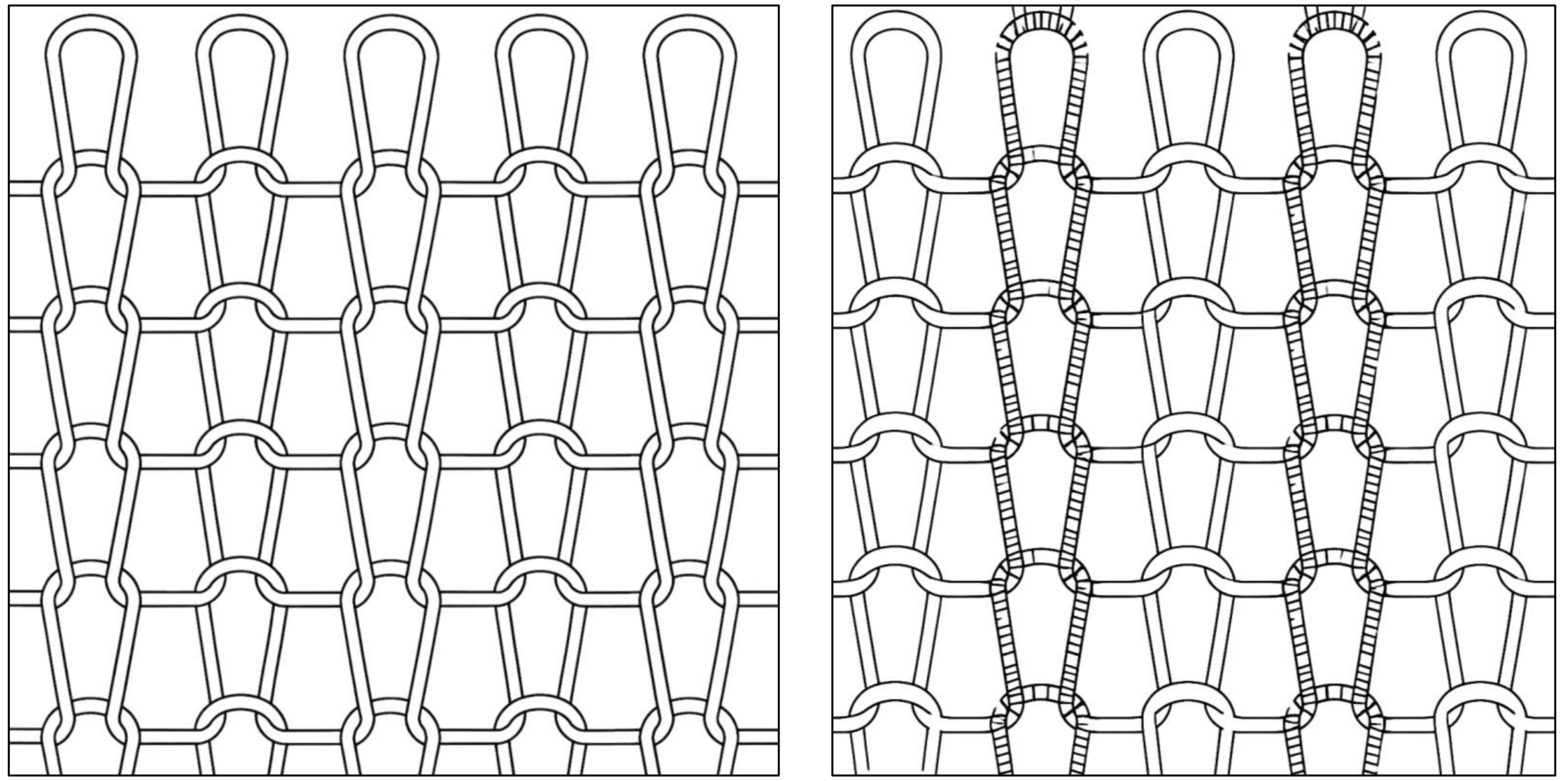
5. Laddering

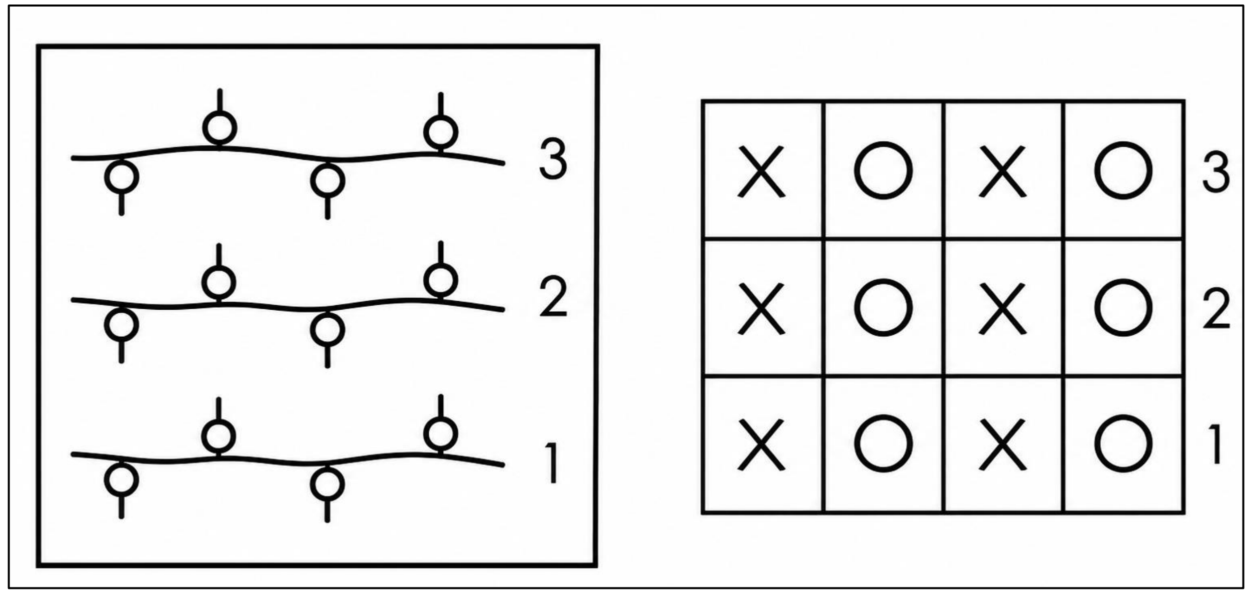
- A dropped stitch can cause a ladder to form.
- Laddering is higher with: Smooth yarns, Large loops, Stretched fabric
- In rib fabric, laddering only moves toward the first knitted edge because the connecting loops prevent it from spreading in the opposite direction.

6. Weight and Thickness

- Rib fabric is thicker, heavier, and bulkier than plain knit fabric made from the same yarn.
- A 1×1 rib fabric is about twice as thick and heavy as plain knit fabric.
- After removing from the machine, rib fabric becomes narrower and thicker because the front loops cover the back loops.

DRAW THE LOOPING DIAGRAM OF TECHNICAL FACE AND TECHNICAL BACK OF A RIB FABRIC WITH NOTATION





HOW TO DISTINGUISH A RIB FABRIC?

- Appearance: Same on both sides. Like face of plain.
- Unroving: Only from end knitted last.
- Curling: No tendency to curl.
- Thickness & Warmth: Much thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Moderate – Very High – High.
- End uses: Socks, Cuffs, Waistbands, Collar, Men’s outwear, Underwear.

FEATURES OF PURL FABRIC

- Purl fabric is composed of sequentially 1 course back and 1 course wales that a horizontal or course-wise effect is produced.
- Both face and back loops remain in the same wales. The face loops are covered so the needle and connection arches typical of reverse loops show on both sides. So, the fabric imitates back side of single jersey.
- Needle can change the needle bed in alternate cycle. Same needle produces face loop on one bed and back loop on other bed. Needle slider is used to move the needle set from one bed to another.
- This fabric has no tendency to curl when it is cut.
- It can be pulled out from either end.
- Double ended latch needles are used to produce purl fabric.
- Its thickness is theoretically double to that of a plain knit fabric.

FEATURES OF PURL FABRIC

- It is a balanced structure and has a soft handle.
- Purl structure creates more extensibility in length wise direction. This is an advantage, especially when compared with the very limited lengthwise extensibility of plain or rib structures. The fabric is stretchable in the width as with all loop-based structures.
- Purl knit structures are best suited to infant's wear, socks, golf sweaters and sportswear. This flexibility in length and width makes the purl knit ideal for baby wear where elongation and expansion are required due to the fast-growing rate of infants and to simplify the dressing process.

FEATURES OF PURL FABRIC

1. Appearance

- In purl fabric, each wale and course can contain both face and reverse loops.
- The fabric contracts in the length direction.
- The face loops are hidden, and the reverse loop arches appear on both sides.
- It is called "Links-Links" (Reverse-Reverse) because both sides have a similar reverse appearance.

2. Extensibility (Stretchability)

- Purl fabric is elastic in the length direction due to its contraction.
- It can also stretch in the width direction like other knitted fabrics.
- Its good length and width flexibility make it suitable for baby wear.

FEATURES OF PURL FABRIC

3. Edge Curling

- Purl fabric is usually balanced and does not curl.
- If the arrangement of face and reverse loops is uneven, some curling may occur.

4. Unroving (Unravelling)

- Purl fabric can be easily unravelled from the last knitted edge.
- Basic purl structures (1×1, 2×1) can be unravelled from both ends, similar to plain knit fabric.

5. Laddering

- A dropped stitch can cause a ladder to form.
- In basic purl structures, laddering can occur both upward and downward, like plain knit fabric.

FEATURES OF PURL FABRIC

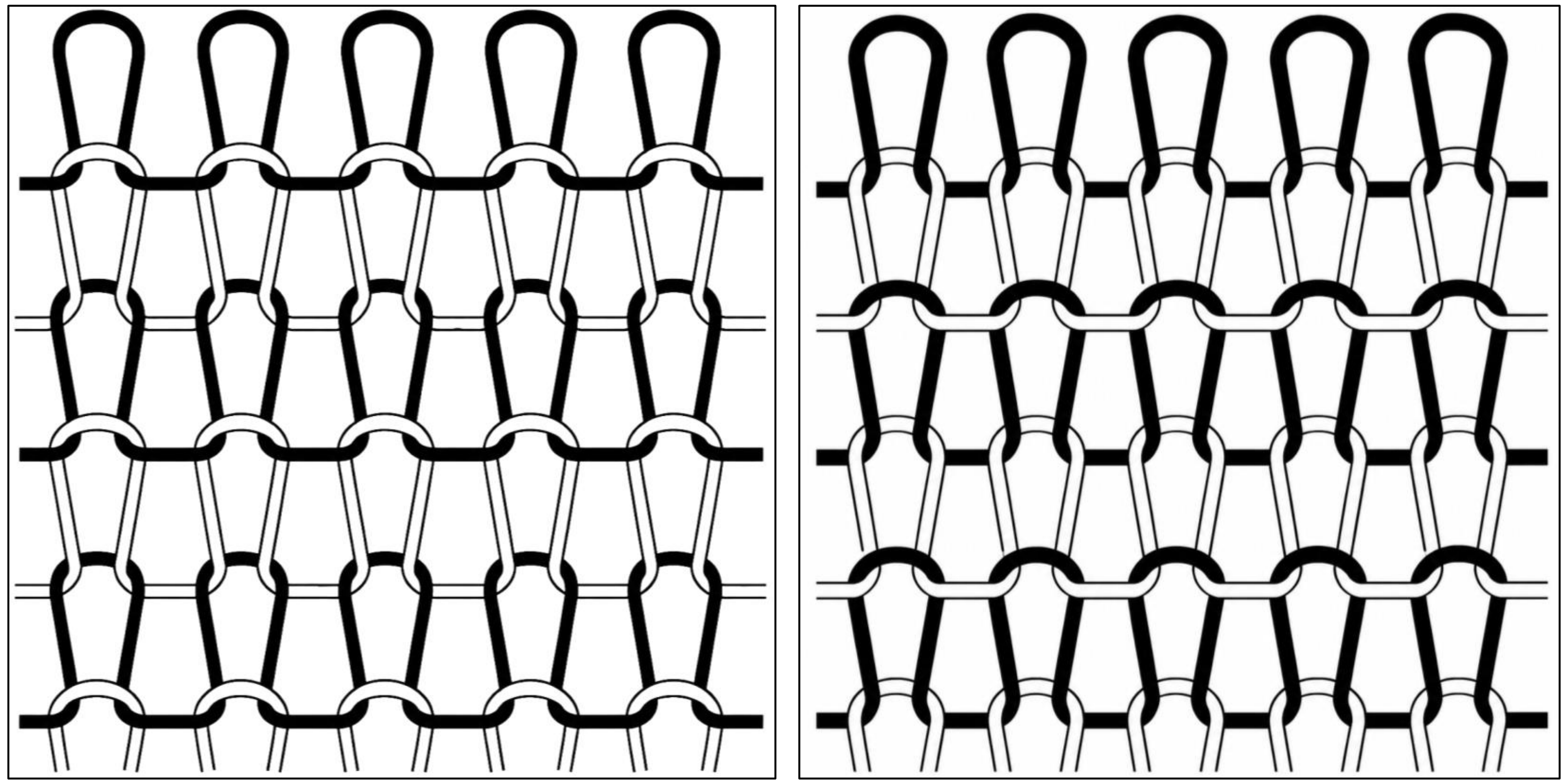
6. Weight and Thickness

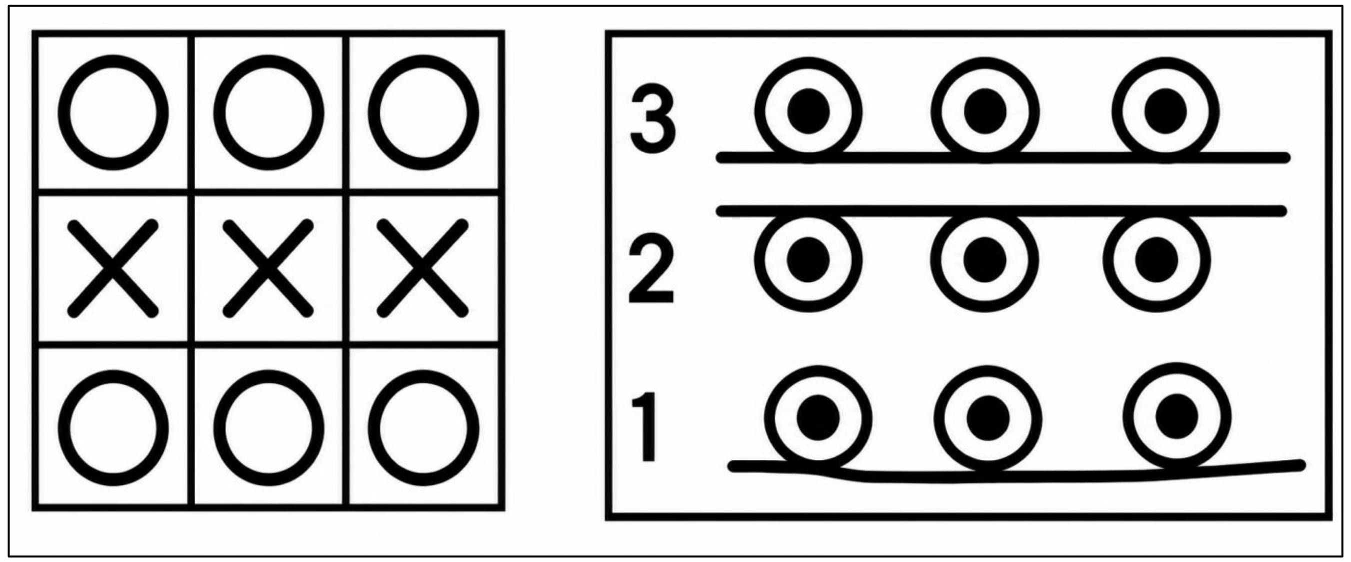
- Purl fabric is soft, bulky, and thicker than plain knit fabric made with the same yarn.
- It has good thermal insulation properties (keeps warmth).

End-use:

Purl fabrics are widely used for baby wear children's clothing, sweater, knitwear, thick and heavy outerwear, under garments etc.

DRAW THE LOOPING DIAGRAM OF A PURL FABRIC WITH NOTATION





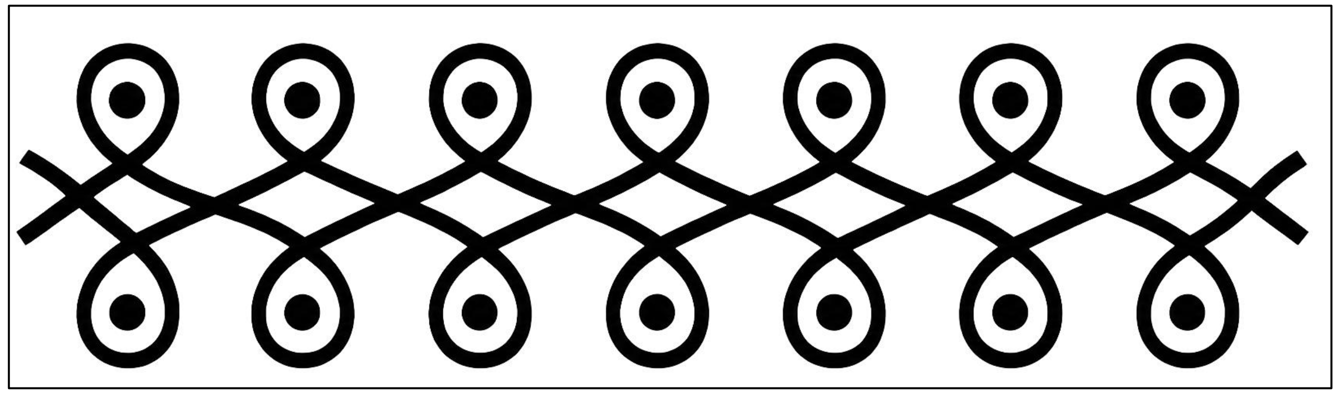
HOW TO DISTINGUISH A PURL FABRIC?

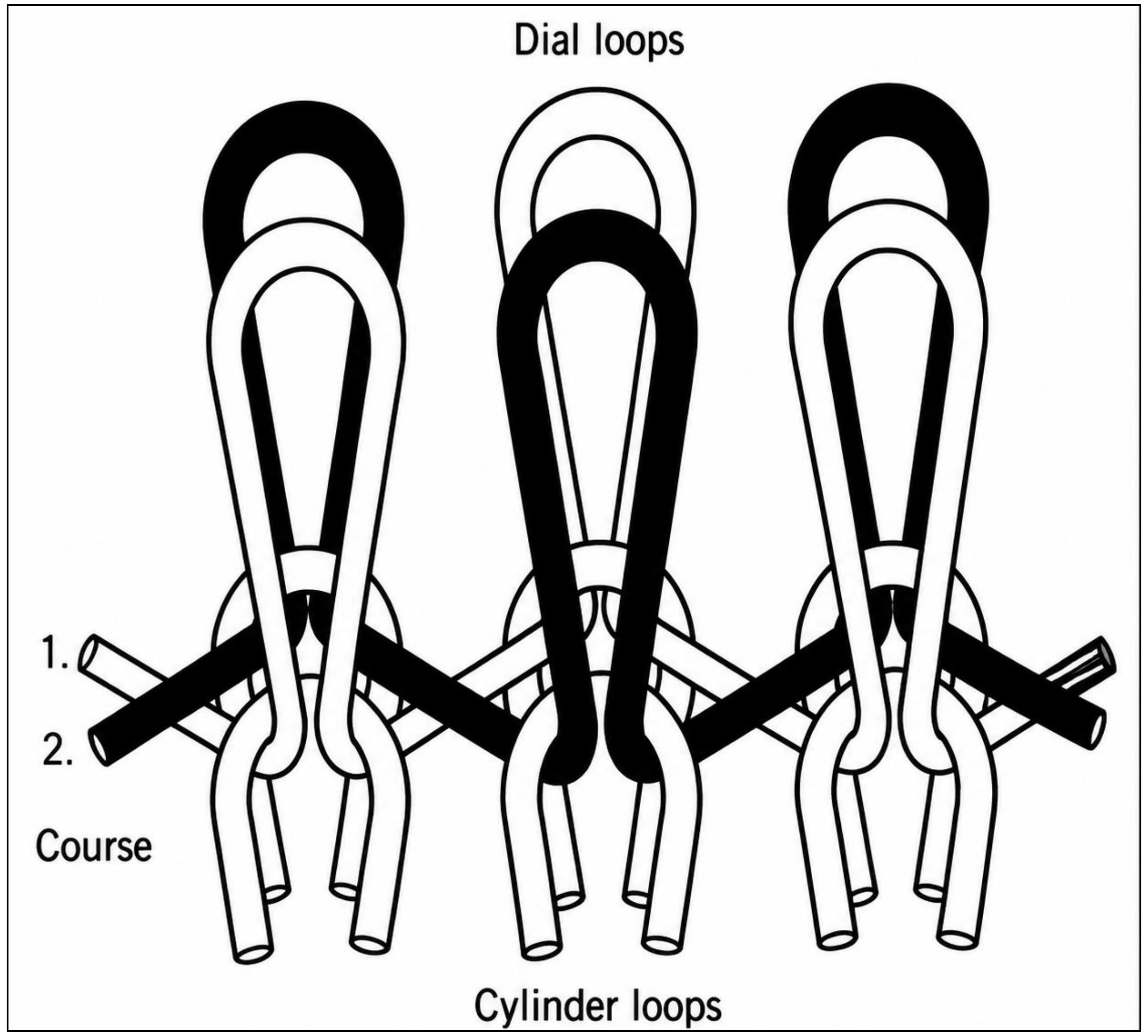
- Appearance: Same on both sides. Like back of plain.
- Unroving: Either end.
- Curling: No tendency to curl.
- Thickness & Warmth: Very much thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Very High – High – Very High.
- End uses: Children's clothing, Thick & heavy outwear.

FEATURES OF INTERLOCK FABRIC

- Interlock has the technical face of plain fabric on both sides.
- It cannot be pulled out from either end. It can only be pulled out from edge knitted last.
- It is a balanced, smooth and stable structure. it is thicker, heavier and narrower than rib of equivalent gauge, and requires a finer, better, more expensive yarn
- It lies flat without curl when cut.
- Face and back of plain interlock fabrics are same. Wales prominent on the both sides. The fabric imitates face side of single jersey fabric.
- The needles in two beds being exactly opposite each other (Face to face gaiting) so that only one of two can knit at one feed. The wales on each side are exactly opposite each other.
- Each interlock pattern row or interlock course requires two feeder courses. So, 1 x 1 interlock structure is produced by two 1 x 1 rib structure interlace to each other. Productivity is half of rib knitting machine.
- Interlocks fabrics are thicker and heavier than rib fabrics.

DRAW THE LOOPING DIAGRAM OF AN INTERLOCK FABRIC WITH NOTATION





HOW TO DISTINGUISH AN INTERLOCK FABRIC?

- Appearance: Same on both sides. Like face of plain.
- Unroving: Only from end knitted last.
- Curling: No tendency to curl.
- Thickness & Warmth: Very much thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Moderate – Moderate – Moderate.
- End uses: Underwear, Shirts, Suits, Trouser, Sportswear.

IDENTIFICATION OF SINGLE JERSEY AND DOUBLE JERSEY FABRIC

Knitted fabrics are mainly divided into single jersey and double jersey depending on whether they are made using one or two sets of needles.

Single Jersey Fabric

- ❑ Made using one set of needles.
- ❑ All face loops appear on one side of the fabric.
- ❑ All back loops appear on the other side.
- ❑ The two sides have different appearances.

IDENTIFICATION OF SINGLE JERSEY AND DOUBLE JERSEY FABRIC

<u>Double Jersey Fabric</u>

- ❑ Made using two sets of needles.
- ❑ Both sides may contain:
    - ❖ Only face loops (rib and interlock fabrics)
    - ❖ Only back loops (purl fabric)
    - ❖ A combination of face and back loops.
- ❑ In basic balanced structures, both sides look the same.
- ❑ In other structures, the two sides may look different.

THANK YOU !